SAFETY DATA SHEET

According to EC 2020/878 (REACH) and 1272/2008 (CLP)

Last update: September 12, 2024

1. IDENTIFICATION OF THE SUBSTANCE/ MIXTURE AND OF THE COMPANY/ UNDERTAKING

1.1 PRODUCT IDENTIFIER

Product name: Filament Noctuo Cosmic

1.2 RELEVANT IDENTIFIED USES OF THE SUBSTANCE OR MIXTURE AND USES ADVISED AGAINST

Intended use: FDM 3D printing material

1.3 DETAILS OF THE SUPPLIER OF THE SAFETY DATA SHEET

Supplier:	**Manufacturer:**
Noctuo Sp. zo.o. ul. Sowińskiego 5 44-121 Gliwice, Polska	Noctuo Sp. zo.o. ul. Sowińskiego 5 44-121 Gliwice, Polska

1.4 EMERGENCY TELEPHONE NUMBER

Emergency telephone: +48 42 63 14 724

2. HAZARDS IDENTIFICATION

2.1 CLASSIFICATION OF THE SUBSTANCE OR MIXTURE

Classification according to Regulation (EC) N° 1272/2008 (CLP): Not classified as hazardous (polymeric state)

2.2 LABEL ELEMENTS

The product does not require labelling according to GHS criteria in accordance with Regulation (EC) No 1272/2008 of the European Parliament and of the Council.

Hazard pictograms: Not applicable

Signal word: Not applicable

Hazard statement: Not applicable

2.3 OTHER HAZARDS

Not applicable.

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1 SUBSTANCE

3.2 PREPARATION

Nr CAS	Chemical name	Impurities Contributing to Hazard
9051-89-2	Polylactide	None

This material may generate dust that cannot be classified (PNOC). According to the Occupational Safety and Health Administration (OSHA) regulations, the permissible exposure limit/time-weighted average for PNOC dust is 15 mg/m3 for total dust and 5 mg/m3 for the respirable fraction. According to the American Conference of Governmental Occupational Health Inspectors (ACGIH) regulations, the threshold/time-weighted average for dust is 10 mg/m3 for respirable dust and 3 mg/m3 for respirable dust.

4. FIRST AID MEASURES

4.1 DESCRIPTION OF FIRST AID MEASURES

General notes:

Special measures are not necessary.

Following eye contact:: Flush with plenty of water for at least 15 minutes. Remove contact lenses. Seek medical advice if necessary.

Following skin contact: In the event of a burn caused by melted product, cool the burnt area immediately, preferably under running water. DO NOT remove the material from the skin. Removal may cause serious tissue damage. Seek medical advice.

Following inhalation: A person exposed to inhalation of volatile compounds of decomposition should be

removed from the contaminated atmosphere, provided with a supply of fresh air, protected from heat loss; seek medical advice if necessary.

Following ingestion: Preventive - rinse mouth with water and drink plenty of water. Do not induce vomiting! Consult a doctor.

4.2 MOST IMPORTANT SYMPTOMS & EFFECTS BOTH ACUTE & DELAYED

No data available.

4.3 INDICATION OF ANY IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENT NEEDED

No data available.

5. FIRE - FIGHTING MEASURES

5.1 EXTINGUISHING MEDIA

Suitable extinguishing agents: Foam. Water. Carbon dioxide (CO_2). Dry chemical. Alcohol-resistant foam is preferred. If alcohol-resistant foam is not available, use regular foam.

Unsuitable extinguishing agents: Not applicable.

5.2 SPECIAL HAZARDS ARISING FROM THE SUBSTANCE OR MIXTURE:

Hazardous decomposition products: Combustion releases toxic fumes: Aldehydes, Carbon monoxide (CO), Carbon dioxide (CO_2).

5.3 ADVICE FOR FIREFIGHTERS

Special Protective Equipment for Firefighters: As in any fire, wear appropriate self-contained, pressure-fed breathing apparatus, MSHA/NIOSH approved or equivalent, and full protective gear.

Firefighting Procedures: Cool containers/tanks with water spray. Water fog may be used to cool closed containers. Dust deposits must not be allowed to accumulate on surfaces, as they may form an explosive mixture if released into the atmosphere in sufficiently high concentration. Avoid dispersing dust into the air (i.e., cleaning dusty surfaces with compressed air).

6. ACCIDENTAL RELEASE MEASURES

6.1 PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT & EMERGENCY PROCEDURES

Use personal protective equipment. Avoid contact with skin and eyes. Avoid dust accumulation. Remove potential sources of ignition. Keep unprotected persons away.

6.2 ENVIRONMENTAL PRECAUTIONS

Do not allow product to reach sewage system or water bodies. Inform respective authorities in case product reaches water, sewage system or soil.

6.3 METHODS AND MATERIAL FOR CONTAINMENT AND CLEANING UP

Remove all sources of ignition. Use explosion-proof equipment. Dust may present a fire or explosion hazard. Vacuum or sweep up material and place in waste container. Keep in closed and appropriate containers for disposal.

6.4 REFERENCE TO OTHER SECTIONS

For information on safe handling, see Section 7.
For information on personal protective equipment, see Section 8.
For information on disposal of contaminated material, see Section 13.

7. HANDLING AND STORAGE

7.1 PRECAUTIONS FOR SAFE HANDLING

Use personal protective equipment as required. Avoid skin and eye contact. Low risk under normal industrial or commercial use. After contact with molten product, quickly cool skin exposure with cold water. Do not attempt to remove solidified material from skin. Do not remove clothing if adhered to skin. Avoid dust formation. May create flammable dust concentrations in air if fine particles are generated during further processing, handling or otherwise.

7.2 CONDITIONS FOR SAFE STORAGE, INCLUDING ANY INCOMPATIBILITIES

Store at a temperature not exceeding 50 °C. Store
in a cool place. No special restrictions for storage with other products.

7.3 SPECIFIC END USE(S)

The product is designed for use as a filament for 3D printing using FDM technology.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 CONTROL PARAMETERS

Provide adequate exhaust ventilation.

8.2 EXPOSURE CONTROL

Eye protection:	Not required if dust or fumes are not generated. If present, wear safety glasses.
Skin and body protection:	Impermeable clothing.
Respiratory protection:	Under normal circumstances, filtering face masks should be available for use when necessary.
Hand protection:	Protective gloves, if necessary.
Other:	Protection against contact with molten product.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 INFORMATION ON BASIC PHYSICAL AND CHEMICAL PROPERTIES

Physical state	Solid
Appearance:	Circular profile
Odour:	Sweet
pH:	Not applicable
Melting point / freezing point:	150 – 180 °C
Initial boiling point and boiling range:	Not applicable
Evaporation rate:	Not specified
Flammability (solid, gas)	Yes
Vapour pressure:	Not specified
Vapour density:	Not specified
Density:	1.25 g/cm3
Solubility(ies):	Insoluble in water
Auto-ignition temperature:	388 °C
Decomposition temperature:	250 °C

9.2 OTHER SAFETY INFORMATION

None.

10. STABILITY AND REACTIVITY

10.1 REACTIVITY

Under normal use conditions none.

10.2 CHEMICAL STABILITY

Stable under recommended conditions.

10.3 POSSIBLE HAZARDOUS REACTION

Not applicable.

10.4 CONDITIONS TO AVOID

Temperatures above 230°C. Avoid prolonged melting and storage for long periods of time at very high temperatures.

10.5 INCOMPATIBLE MATERIALS

Oxidizers, strong bases.

10.6 HAZARDOUS DECOMPOSITION PRODUCTS

Combustion produces odorous and toxic fumes, aldehydes, carbon monoxide (CO), carbon dioxide (CO2).

11. TOXICOLOGICAL INFORMATION

11.1 INFORMATION ON TOXICOLOGICAL EFFECTS

Caustic effect:	Based on available data, the classification criteria are not met.
Mutagenic and carcinogenic effects:	Not mutagenic in the AMES test, does not contain carcinogenic or potentially carcinogenic substances listed by OSHA, IARC or NTP

Sensitizing effects on the respiratory tract:	Based on available data, the classification criteria are not met.
Sensitizing effects on the skin:	Based on available data, the classification criteria are not met.
Serious eye damage, eye irritation:	Based on available data, the classification criteria are not met.

11.2 INFORMATION ON OTHER HAZARDS

Local effects: Vapours during printing may irritate the eyes, skin and respiratory system. Resin particles, like other inert materials, cause mechanical eye irritation. Ingestion may irritate the digestive system, cause nausea, vomiting and diarrhea.

12. ECOLOGICAL INFORMATION

12.1 TOXICITY

EC50/72h/algae > 1100 mg/l

12.2 PERSISTENCE AND DEGRADABILITY

No data available.

12.3 BIOACCUMULATIVE POTENTIAL

It does not accumulate in living organisms.

12.4 MOBILITY IN SOIL

No data available.

12.5 RESULTS OF PBT AND VPVB ASSESSMENT

Not applicable.

12.6 OTHER ADVERSE EFFECTS

No data available.

12.7 ADDITIONAL INFORMATION

Use the product in accordance with well-proven working methods, preventing its release into the environment.

13. DISPOSAL CONSIDERATION

13.1 WASTE TREATMENT METHODS

In accordance with local and national regulations (i.e. EU 91/15/CEE, EU 91/689/CEE, EU 94/62/CEE AND the following them) waste should not be released into the environment. Do not pollute ponds, waterways or canals with product or used packaging.

14. TRANSPORT INFORMATION

Not classified as dangerous goods according to transport regulations (IMDG, IATA, ADR / RID).

14.1 UN number

Not applicable

14.2 UN proper shipping name

Proper Shipping Name: NOT REGULATED

14.3 Transport hazard class(es)

Not applicable.

14.4 Packing Group

Not applicable.

14.5 Environmental hazards

Not applicable.

14.6 Special precautions for user

Not applicable.

14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable.

15. REGULATORY INFORMATION

15.1 SAFETY, HEALTH AND ENVIRONMENTAL REGULATIONS /LEGISLATION SPECIFIC FOR THE SUBSTANCE OR MIXTURE

EU regulations; 2015/830/EC, 1272/2008/EC, 1907/2006/EC
Directives; 2006/8/EC, 2004/73/EC, 1999/45/EC, 67/548/EEC

15.2 CHEMICAL SAFETY ASSESSMENT

For this substance a chemical safety assessment is not yet required.

16. OTHER INFORMATION

The information contained in this sheet is based on current knowledge and experience with the product. This information is intended to support independent assessment of the methods used to ensure safe work with the filament and its safe disposal.